

05/10/2026

Continuing on with the analytical thermalization implementation. Have finished implementing the analytical thermalization into it's own little object. It is able to produce the expected system occupation, as well as plot $A(\omega)$. The simulation object generates a thermalization object such that it can compare the analytical vs numeric thermalization on the plot (as in Figure 1, Figure 2)

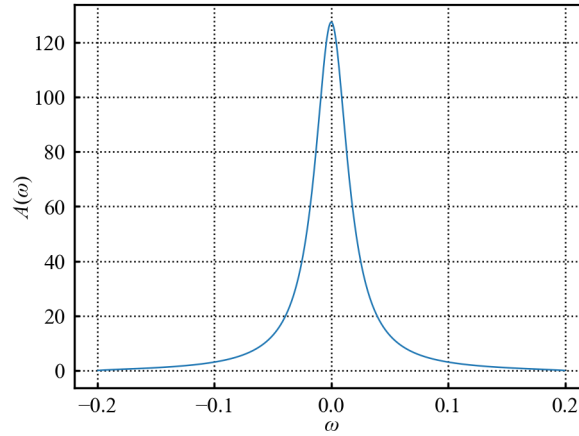


Figure 1: $A(\omega)$ for standard parameters

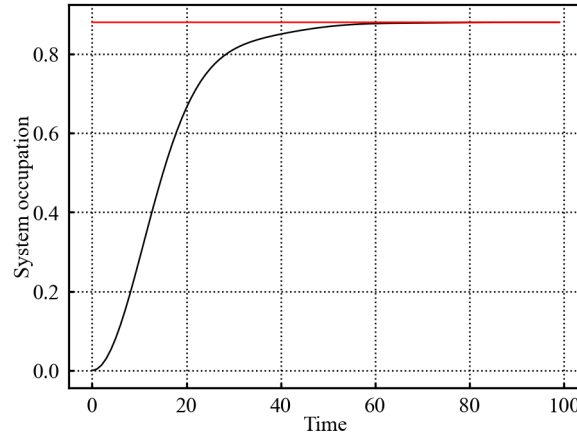


Figure 2: System occupancy over time with red line representing analytical thermalization value

That took significantly longer than I would have liked; trying for symbolic automation of that integral was a poor idea.

Have also quickly setup the bones of the dual simulation system. Idk yet how to extract the dynamical map, but know that it will involve 2sims, 1 w/system starting occupied, the other w/unoccupied system. SO have setup a class that contains and runs and occupied and unoccupied simulation alongside basic utilities for it.

06/10/2026

Starting to look at the dynamical map; David has updated (Strachan 2026) with a quick outline.

Reading notes: Dynamical map extraction (Strachan 2026)

Because the simple model we're using only allows the system to be in a $|0\rangle$, or $|1\rangle$ state, we expect the dynamical map to be a 2x2 matrix of form:

$$\Lambda(t) = \begin{bmatrix} \nu & \sigma \\ \alpha & \beta \end{bmatrix} \quad (1)$$

Considering it's action on some state of occupation p $|\psi\rangle = \begin{bmatrix} 1-p \\ p \end{bmatrix}$ and enforcing trace preservation yields:

$$\Lambda(t) |\psi\rangle = \begin{bmatrix} \nu(1-p) + \sigma p \\ \alpha(1-p) + \beta p \end{bmatrix} \quad (2)$$

$\downarrow$

$$\nu(1-p) + \sigma p + \alpha(1-p) + \beta p = 1$$

$$(\nu + \alpha) + (\sigma + \beta - \nu - \alpha)p = 1$$

which can only hold $\forall p$ if:

$$\Lambda(t) = \begin{bmatrix} \nu & \sigma \\ 1-\nu & 1-\sigma \end{bmatrix}$$

Need to ensure “positivity” (unsure as to exactly why, but will roll w/it). Equation 2 maps initial occupation p to a new occupation $p^* = (1-p) + p = (1-p) + p$. Oh i see why positivity;

p^* represents an occupation probability; negative values are nonsensical ergo $p^* \in [0, 1]$ is a useful condition. Expressing $p^* = C_{11}(t)$ where C is the correlation matrix directly links the dynamical map s.t:

$$C_{11}(t) = \begin{cases} 1 - \nu & C_{11}(0) = 0 = \textit{occupied} \\ \nu - \sigma & C_{11}(0) = 1 = \textit{unoccupied} \end{cases} \quad (3)$$

Therefore:

$$\Lambda(\hat{t}) = \begin{bmatrix} 1 - C_{occ}(t) & 1 - C_{occ}(t) - C_{unocc}(t) \\ C_{occ}(t) & C_{occ}(t) + C_{unocc}(t) \end{bmatrix} \quad (4)$$

where $C_{occ}(t) = C_{11}(t)$ for the initially occupied state, $C_{unocc}(t) = C_{11}(t)$ for the initially unoccupied state.

It is a touch unclear to me over which t $C_{11}(t)$ needs to be measured. My naive assumption is that it needs to be defined over all t of interest; so simulate till thermalization and then $\Lambda(t)$ encapsulates the dynamics over that time range. If so, the dynamical map would probably be stored as either a time-instantaneous dynamical map dictionary, or in an array. Far less storage required compared to the full correlation matrix (besides which the dynamical map is the primary object of analysis for later).

But assuming i'm correct, then implementing the dynamical map should simply be a matter of querying $C_{11}(t)$ for the occupied and unoccupied simulations respectively, and inputting those values into Equation 4.

Have added some empty functions and code comments to the dual sim object; nothing functional yet, just systems that will support this dynamical map extraction and use.

07/10/2026

Had supervisor meeting:

- Explained what i had done thus far
- Briefly discussed presentation:
 - Short explanation of what my project is and what the plan is
 - 1st slide, explain general research area
 - 2nd slide, explain specific problem and questions
 - explain general methods and background
 - show current results
 - outline plan

- Note to self: is to a large degree pedagogical exercise; need to explain topic to unfamiliar peers; if they understand overview of what i’m doing, that’s a win
- Note: can get system occupation from dynamical map; plotting this may be useful gut check to ensure everything is working
- Short-mid term plan:
 - Implement dynamical map into code
 - Implement memory kernel as first measure of non markovianity
 - Do more literature reading - I’ve spent a lot of time implementing a solid technical foundation for the project, but have spent comparatively little time delving into the literature and the theory
 - prep for presentation
- Provided additional papers; *(Cerrillo and Cao 2014) should outline the memory kernel stuff,
 - (Morillas-Rozas et al. 2026) may provide useful comparison of numerical kernel and analytical limit. This latter paper should provide insight into the subtleties and potential pitfalls if i don’t treat this carefully

Consolidating the plan for the foreseeable, i think it follows:

- Short term (pre presentation):
 - Wk of 05/10/2026: Implement dynamical map, implement plotting and analytical utilities for it in order to gut check it generally - maybe polish some of the plotting utilities to make them more flexible for
 - Wk of 12/10/2026: Implement memory kernels and use it as case study to help guide literature deep dive
 - Wk of 19/10/2026: Block week out to work on presentation
 - Wk of 26/10/2026: Presentation slides are due this Wk - can take some time to practice presentation
 - Wk of 02/11/2026: Presentations occur
- Mid-long term (post presentation):
 - Finish off literature deep dive - this should give me a firm conceptual framework for the topic area and the different measures of non markovianity
 - Implement other memory definitions
 - Compare memory definitions on current setup
 - Figure out next steps w/insight gained at this point

Cerrillo, Javier, and Jianshu Cao. 2014. “Non-Markovian Dynamical Maps: Numerical Processing of Open Quantum Trajectories.” *Physical Review Letters* 112 (11): 110401. <https://doi.org/10.1103/PhysRevLett.112.110401>.

Morillas-Rozas, Marcel, Alberto López-García, Gonzalo Reina Rivero, Jianshu Cao, and Javier Cerrillo. 2026. “The Transfer Tensor Method: An Analytical Study Case.” *The Journal of Chemical Physics* 165 (5): 054108. <https://doi.org/10.1063/5.0340784>.

Strachan, David J. 2026. “MarkovianityOutline.”